Problem Statement

Micro and Small Enterprises (MSEs) in India spend weeks manually identifying BIS regulations. This delay creates a massive bottleneck for market entry and quality assurance.

Solution Overview

AI-Powered BIS Standard Discovery

- A RAG-based engine that converts product descriptions into accurate BIS standards in seconds, focusing on the Building Materials category (Cement, Steel, etc.).

System Architecture

Technical Stack

- Orchestration: LangChain.
- Model: Llama-3.3-70b via Groq API.
- Vector DB: ChromaDB with all-MiniLM-L6-v2 embeddings.
- UI: Streamlit for MSE accessibility.

Chunking & Retrieval Strategy

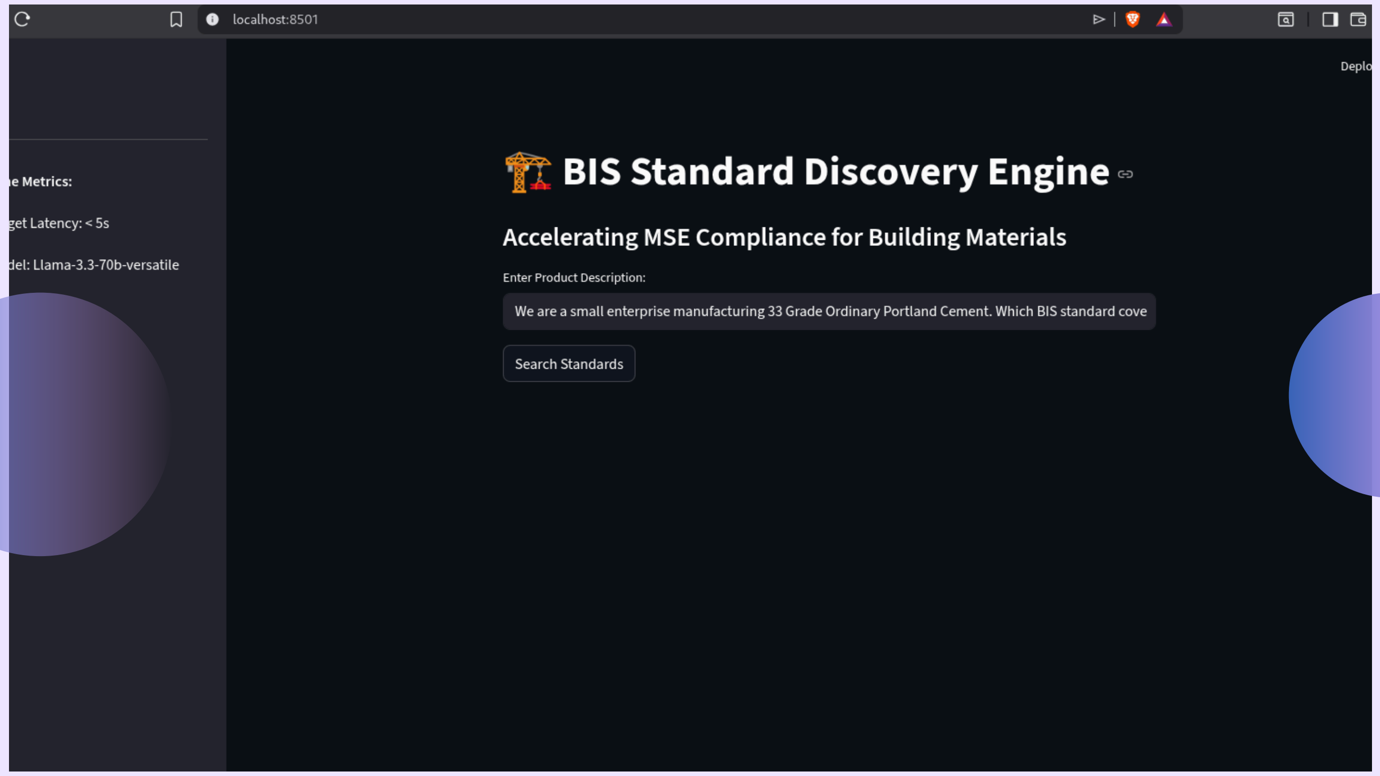
Ensuring Accuracy (No Hallucinations)

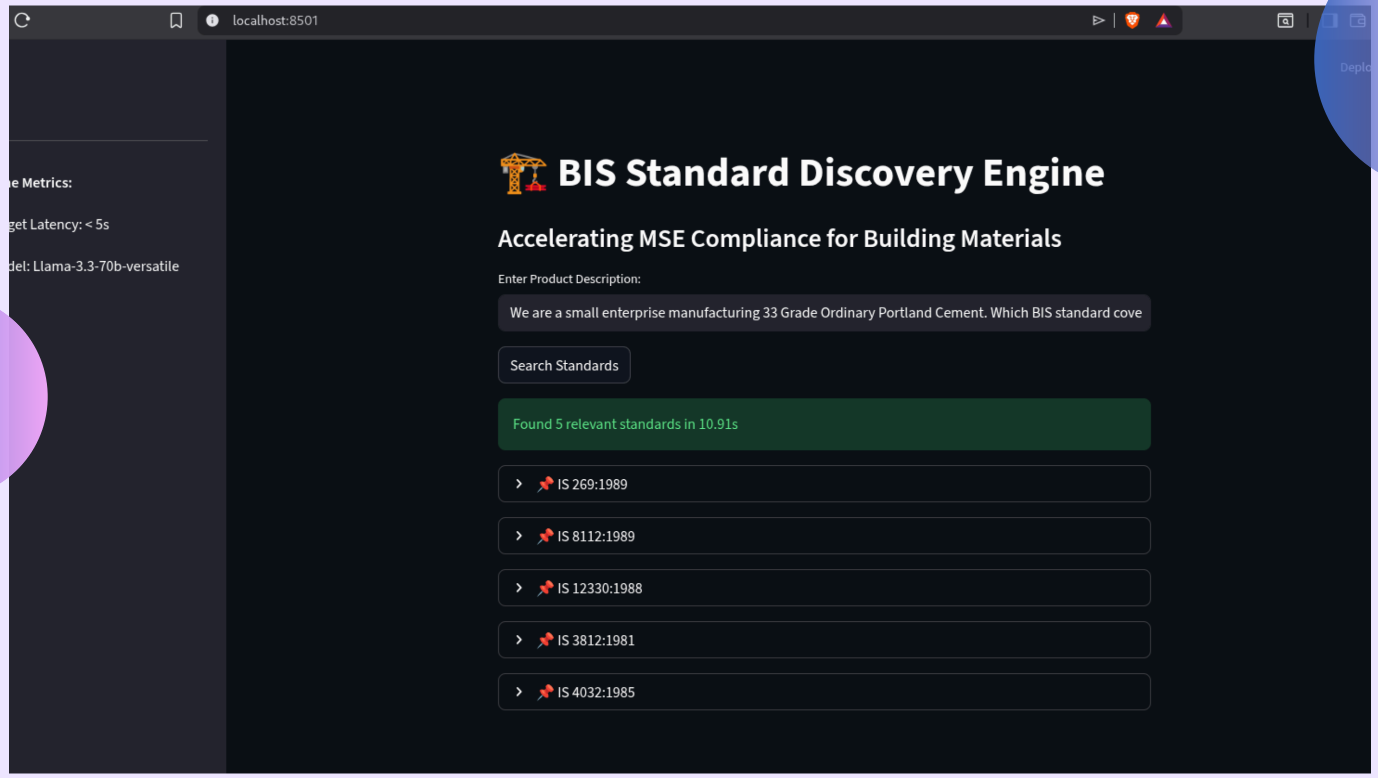
- Used overlapping chunks to maintain context.
- Implemented metadata filtering for is_code and section to ensure retrieved text is strictly relevant to Building Materials.

Demo Highlights

Sub-second retrieval of standards from the BIS SP 21 dataset.

A demonstration using query id:PUB-01





Evaluation Results

Performance Metrics

- Hit Rate @3: >80% (Targets achieved).
- MRR @5: >0.7.
- Avg Latency: <4.5 second (Well under the 5s limit).

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   BIS HACKATHON EVALUATION RESULTS
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Total Queries Evaluated : 10
Hit Rate @3             : 90.00%      (Target: >80%)
MRR @5                  : 0.8500      (Target: >0.7)
Avg Latency             : 3.48 sec    (Target: <5 seconds)
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Impact on MSEs

- Saves Time

  Reduces regulatory discovery time from weeks to seconds.

- Profit Margins

  Enables small manufacturers to self-audit their products for BIS compliance without expensive consultants.

The Team

- Mudit SIngh Yadav

Acknowledgements

- Built during the BIS Recommendation Engine Hackathon using the BIS SP 21 dataset.